

Chapter 1

Introduction to Collaboration

This chapter covers the following topics:

Audio Communication: This topic will illustrate a brief history of audio communication as it has evolved and provide a high-level overview of audio communication that exists today.

Video Communication: This topic will illustrate a brief history of video communication as it has evolved and provide a high-level overview of video communication that exists today.

Unified Communication: This topic will identify different forms of communication and illustrate how these means of interaction can be leveraged in today's business enterprise.

Driving Change in the Industry: This topic will illustrate how the industry has transformed over the last 20 years and speculate where the industry is heading in the future.

What comes to mind when you hear, or in this case read, the word *collaboration*? To collaborate means to work together with someone else, usually to achieve an end goal or produce a product. Communication today requires so much more than just having a voice conversation with someone else. Cisco has taken on the vision to provide the best products that will help employees within businesses truly collaborate. This is why Cisco, and many other vendors, have embraced the term *collaboration* to replace *communication*. This chapter will provide a history of voice and video communication and illustrate the evolutionary path communication has followed leading to the collaboration architecture that exists today. Topics discussed throughout this chapter include the following:

- [Audio Communication](#)
- [Video Communication](#)
- [Unified Communication](#)
- [Driving Change in the Industry](#)

“Do I Know This Already?” Quiz

The “Do I Know This Already?” quiz allows you to assess whether you should read this entire chapter thoroughly or jump to the “Exam Preparation Tasks” section. If you are in doubt about your answers to these questions or your own assessment of your knowledge of the topics, read the entire chapter. [Table 1-1](#) lists the major headings in this chapter and their corresponding “Do I Know This Already?” quiz questions. You can find the answers in [Appendix A](#), “[Answers to the ‘Do I Know This Already?’ Quizzes.](#)”

Table 1-1 “Do I Know This Already?” Section-to-Question Mapping

Foundation Topics Section	Questions
Audio Communication	1–2
Video Communication	3–4
Unified Communication	5
Driving Change in the Industry	6–7

Caution

The goal of self-assessment is to gauge your mastery of the topics in this chapter. If you do not know the answer to a question or are only partially sure of the answer, you should mark that question as wrong for purposes of the self-assessment. Giving yourself credit for an answer you correctly guess skews your self-assessment results and might provide you with a false sense of security.

- 1.** What solution allows businesses to place and receive analog voice calls over a circuit-switched network?
 - a.** PBX
 - b.** ISDN
 - c.** POTS
 - d.** VoIP
- 2.** What solution allows businesses to place and receive digital voice calls over a packet-switched network?
 - a.** PBX
 - b.** ISDN
 - c.** POTS
 - d.** VoIP
- 3.** What ISDN standard did the first video endpoints use to communicate?
 - a.** POTS
 - b.** H.320
 - c.** H.323
 - d.** SIP
- 4.** Which of the following are considered a Telepresence endpoint?
 - a.** Immersive video system
 - b.** Conference room video system
 - c.** Personal desktop video system
 - d.** All of these answers are correct
- 5.** Which of the following modes of communication allows real-time text messages to be sent and received?
 - a.** Email
 - b.** IM

c. Presence

d. Fax

6. Which of the following protocols allows RTP and RTCP streams to be multiplexed together?

a. TIP

b. SIP

c. H.323

d. H.320

7. Which of the following Cisco products uses cognitive collaboration to enhance the user experience?

a. CUCM

b. CMS

c. CTS3000

d. Webex

Foundation Topics

Audio Communication

What is audio communication? Audio communication is simply information being transmitted in a form that can be heard. Two people talking together over a cup of coffee is audio communication. It can be any form of sound from speech to music. Rhythm alone can be communication. Ancient tribes would use bird calls and animal sounds to communicate without giving away their position. Morse code is even a form of audio communication. Of course, audible sound for the purpose of communication isn't restricted to just humans; it's all over the natural world. For the purpose of this definition, it is important to differentiate audio communication from other audible sounds based on intent. Be it

animal or human, when an audible sound is intended to be recognized and understood, audio communication is occurring.

Fast forward from the bird calls of ancient times, humankind has sought to achieve communication over vast distances for many years. Pigeons and other animals have even been used to send messages. Then the discovery of how to use electricity propelled us into a new age of invention and revolution. In the 1830s a man named Samuel Morse created the telegraph system. This system worked by sending electrical pulses over a wire between stations. When those electrical pulses were received, they would transmit a tone, which was sent in a pattern. The code that was created to map those tones to written language is known as Morse code, and it is still used today.

The telegraph incited creative thinkers to imagine ways actual audio voice could be sent over that same electrical wire. This vision was made a reality in 1849 when a man named Antonio Meucci invented the first basic telephone. From that time there were others who invented different versions of the telephone, but it was Alexander Graham Bell who won the first U.S. patent for the telephone in 1876. The telephone system continued to evolve from there. The first manual telephone exchange opened in 1878.



Operators would take incoming calls and manually connect them through to the intended destination. This system worked for a short while, but there was an unexpected interruption to this system known as the common cold. These operators had to be trained to perform their jobs, so when they got sick and couldn't come into work, finding a replacement was no simple task. In 1889, the first automatic telephone exchange was invented, and over the next few years this system would replace the manual telephone exchanges.

This phone system continued to operate for nearly 100 years before the next big revolution would occur with the telephony system. In 1986, a new set of protocols was introduced that allowed the traditional analog phone system to be replaced with a digital phone system known as the Integrated Services Digital Network (ISDN). A more detailed explanation of analog and digital signaling will be provided in [Chapter 2](#), "[Audio Basics](#)," and a detailed

explanation of ISDN will be provided in [Chapter 5, “Communication Protocols”](#); however, a brief explanation of these different types of communication should be understood at this time. A plain old telephone service (POTS) is the means of audible communication using analog signals transmitted over wire. ISDN is similar to POTS, but the original analog signal is converted to digital format before it is sent across a wire. That digital signal must be converted back to analog at the receiving phone.

Technically, cellular communication preceded ISDN. The first cellular call was placed in 1973; however, cell phones were very inefficient, heavy, and very expensive. It was ten years from that first cellular call before cell phones went on sale in the United States. In 1983, you could buy a cell phone for \$4000 USD, the cellular network was limited, and the phones were still too big and heavy to fit in your pocket, or even wear on a sexy clip attached to your belt. The cellular industry didn’t start taking off until the mid- to late-1990s. Today cellular technology is very common. It allows audio communication to be transmitted wirelessly across a given area, or cell. These cell towers are very prominent across the globe, enabling people to place and receive calls from virtually any location on the planet, with some obvious limitations. Most modern cell phones also support data connections, opening up many other channels of communication through social media applications and other communication tools, but more on that later.

Before I go into the next advancement in audio communication, an explanation is warranted for another device introduced to business communication systems. This device is known as the private branch exchange (PBX). A PBX operates in a similar fashion to the automatic telephone exchange, except that the purpose of a PBX is to route calls within a business exclusively. PBXs can also connect to the outside world over the public telephone network, but it operates on the same circuit-switched network using POTS or ISDN. In 1997, a company called Selsius Systems released one of the first IP PBXs that supported voice over Internet Protocol (VoIP). Recognizing what that company had created, Cisco quickly swooped it up, and by November 1998, Cisco had acquired Selsius. The original IP PBX was named Call Manager, and this is the same product, though a few upgrades later, that we know as the Cisco Unified Communications Manager (CUCM).



VoIP is an audio technology that allows voice communication to be translated into a binary format and encapsulated into packets and sent across an IP network. This type of communication is known as *packet switched*. Engineers began experimenting with VoIP technologies as early as the 1980s; however, limitations in network bandwidth and audio compression algorithms were too restrictive to develop a product to go to market. By the mid 1990s, the introduction of broadband and high-speed Internet made VoIP a reality beyond what any could have imagined before.

Video Communication

Now that audio communication has been defined, and the history of how audio communication evolved has been scrutinized, the next form of communication to examine is video communication. So, what is visual communication? Visual communication is a method of communicating through sight. Different types of visual communication could include theater, dance, mime, flag signal waving, sign language, hand signals from a bike or car such as flipping someone the bird, and so on. One of the most common forms of visual communication that people use every day is body language. Proper etiquette teaches how to use body language to better communicate with other people, such as maintaining eye contact, nodding to communicate that the spoken message has been heard, or raising a hand to identify to the speaker that you have a question. Visual communication could also be a shrug of the shoulders or a facial expression. Crossing of the arms suggests the listener is closed off to the message being communicated, but don't mention that to your spouse.

I have taught many Cisco classes, both on-site and remote over a medium such as Cisco Webex. Most students have identified a preference to on-site classes because the environment is more conducive to learning. Even from an instructor perspective, an on-site class is preferred over a Webex class. The reason has everything to do with body language. While teaching, the instructor can watch the students for visual clues as to whether a concept is being understood. If students have confused looks on their faces, then the instructor can rephrase the explanation or ask probing questions to identify

where the point of confusion occurred. I have been known to stop teaching and break for lunch because it was visibly obvious that the students needed sustenance before they were going to absorb any more information. The same has been true at the end of a long day, that the students' minds were suffering from information overload. So, class ended to allow the mind to rest and the information to soak in and resonate before starting again the next day. Remote classes have their advantages too, but the point here is centered around the importance of visual communication. Have you ever listened to a sporting event over the radio? Although the sports announcers do a pretty good job of providing a play-by-play account of what is actually going on during the event, an audible interpretation is not the same as seeing the event for yourself. There is so much more information that can be gathered through seeing than by simply hearing. So, as the adage goes, "a picture is worth a thousand words."

Technologically speaking, a lot of different video communication devices are available on the market today. A walk through Times Square in New York City is a perfect example. This part of the city is literally lit up with the lights from digital signs that encompass the whole of Times Square. Information is communicated through advertisements, stock information, and live TV. Some of the signs are even interactive with the crowds of people meandering through the streets. Other common forms of communicating through video today include television, YouTube video clips, and other social media applications. Recent years have given rise to a new way of communicating over video, which involves calling over a video phone. This form of video communication has become known as video telepresence.

Video telepresence allows participants at different locations to see visual cues while audio communication accompanies the communication. Video telecommunications is a relatively young industry. It was first conceived as early as 1870, but the technology to support the idea would not develop for another 50 years. With the invention of the television, the idea of video communication began to develop. The earliest video telephony systems were simply an analog TV and camera set up at two locations. A lot of other special equipment accompanied this setup, making it very expensive to deploy and use. Although many companies contributed to the development of this industry, two companies stand out above the rest.

The first worth mentioning was a Norwegian company known as Tandberg. Based in Oslo, Norway, Tandberg first incorporated in 1933. The company's main products were radios, televisions, and reel-to-reel tape recorders. After filing for bankruptcy in 1978, the company split off into two distinct organizations. Tandberg Data continued to market and sell the same products to consumers. The original Tandberg company changed its focus to telepresence communications. Tandberg launched its first telepresence endpoint in 1989, which used the ISDN standard H.320 as the medium to communicate. By 1995, Tandberg opened its first office in the United States. After making a key acquisition in 1999, Internet Technology AS, by 2000 Tandberg had launched its first video telepresence endpoint that could support video calling over ISDN or IP. The protocol used for IP communication at this time was the ITU-T standard H.323. The year 2004 was big for Tandberg, which acquired a company called Ridgeway Systems, providing the company with firewall traversal capabilities not yet seen in the industry. Tandberg also released a new endpoint product line that year called the MXP series. By 2005, Tandberg incorporated a new IP protocol into its endpoints for IP communication known as Session Initiation Protocol (SIP). As history will tell you, SIP has become the preferred protocol for use in today's networked environments. Between key acquisitions and superb product development, Tandberg forged a path to become the industry leader in video telepresence until 2010, when Cisco acquired Tandberg. Building on the foundation laid by Tandberg, Cisco has held the reigning title as the industry leader in voice and video communication across the world to date.

Another company that is worth mentioning was called PictureTel Corporation. Founded in the United States in 1984, PictureTel quickly grew to become the telepresence industry leader in the early days of video communication. The company's first ISDN-based video endpoint was released in 1986. By 1990, two of PictureTel's lead engineers broke off to form a new company called Polycom. PictureTel was later acquired by Polycom in 2001. The reason PictureTel is worth mentioning has to do with the contributions the company made to jumpstart this new industry. The company got its start based on a 1981 MIT research paper combining block-based motion compensation and transform coding. This idea of block-based motion compensation and transform coding is what most of the

video compression codecs used today are based on. Without that contribution, who knows where the world of video telepresence would be.



In today's market, many different types of video systems allow a variety of means to communicate. Software clients can be downloaded to computers, tablets, and smartphones, so video communication can be used while on the go. Immersive Telepresence systems can be used to enhance the virtual session for a face-to-face in-person experience. Personal video systems, meeting room video systems, and varying vertical solutions are used by businesses worldwide. Video telepresence allows for business-to-business (B2B) and business-to-consumer (B2C) communication. Not only does visual communication in technology allow participants to see one another, but content also can be shared between devices so that participants at separate locations can see the same documents, drawings, presentation slides, or any other content being shared.

Unified Communication



As was mentioned before, there are more ways to communicate in today's world than just audio and video. *Instant messaging (IM)* is a term that became relevant in 1990. IM is a text-based communications tool that allows real-time text messages to be sent over the Internet. IM was made popular by industry giants such as Microsoft Messenger, Yahoo!, AOL AIM, and others like them. A communications tool that often accompanies IM is known as presence. Presence is an indicator light that identifies if a user is online, offline, away, or busy.

Non-real-time communication services are also used today, such as email. Although email was technically first used in the mid-1970s, it wasn't commonly used until the late 1980s and early 1990s when the Internet was more widely accessible. Fax machines are not commonly used anymore, but on occasion you will still find some businesses that use them to share

documents. Government offices tend to be slower to adapt new technologies, so fax machines are still often used in these types of establishments. Some banking institutions and credit unions will use fax machines to share signed legal documents. Voicemail is another common non-real-time communication service. Voicemail refers to a computer-based system that allows a user to record a personal greeting, which will be played when audio calls are not answered within a predefined amount of time. After the greeting plays, the caller has the option to leave a voice message, which can be played back at the main user's discretion.

Couple audio and video with presence, instant messaging, and persistent chat, and there are a lot of ways people can communicate with one another. Plus, there are other forms of communication not mentioned here, such as social media applications. Compound all of this with vendor autonomy and proprietary protocols, and the world of communication becomes quite complex. The answer to that complexity is Unified Communications (UC).



UC is not necessarily a single product, nor a single manufacturer, but more like a set of products that provides a consistent unified user interface and experience across multiple devices and media types. Ideally, it would be better to use a single vendor's products for an enterprise UC solution, but realistically that is not always possible. In its broadest sense, UC can encompass all forms of communications that are exchanged via the medium of the TCP/IP network. UC allows an individual to send a message on one medium and receive the same communication on another medium. A good example of this is when someone receives a voicemail message; the person receiving the message can choose to access it through email or hear an audio playback through a phone. For those of you who are fans of the TV sitcom *The Office*, there is a good example of a UC product on that show called WUPHF. The character Ryan Howard created a software application called WUPHF that would take incoming communication over any media and would resend it out to the intended target as a Facebook message, Tweet, IM chat, SMS text message, fax, and phone call. As humorous as the concept was in the show, the idea behind it is great. UC is the ability to unify multiple modes of communication under a single platform. This is

what Cisco has strived for over many years and what the company has brought to fruition in multiple product platforms available today.

Driving Change in the Industry

Telepresence is the way of the future. As a technology, video communication has been around for approximately 40 to 50 years; however, the growth of video conferencing has depended heavily on the availability to run on a reliable digital communication network. In the 1980s, when ISDN standards were first introduced, the race began to build and deliver the first video telecommunications device. PictureTel was the clear front-runner in this market, but over a short period that title would change hands several times. Polycom began to pick up steam by creating several proprietary audio compression codecs that would later be adopted as standards. Wielding this superior grasp on the technology, and by acquiring some key companies, Polycom quickly propelled itself to become an industry leader in the video communications market.

There was a big drawback to video communications devices at this time. The technology was new and still being developed. Most system components had to be invented so that the systems would perform in a functional manner. Plus the demand for a video phone was very low. These factors made purchasing these video phones very expensive. Then factor in the fact that ISDN was still relatively new itself, so placing a video call over ISDN was quite expensive as well. All of this changed in the late 1990s and early 2000s with the introduction of broadband and high-speed Internet. Now the race was not to simply come out with a video endpoint, but to come out with an endpoint that was capable of calling over IP, a much less expensive medium for communication. Tandberg was quick to respond with an endpoint that used the H.323 standard for IP communication. Other companies came into the market as well, such as Sony, HP, Radvision, VTEL, and Aethra. Other companies were in the VoIP game, such as Cisco, Siemens, Vonage, and Avaya. As the race for delivering video communications over IP continued, Tandberg made a strategic addition to its systems, allowing IP communications over the IETF SIP protocol. After this newish protocol introduction to the system, the company developed a

superior line of endpoints and began to give Polycom a run for its money as the video communications front runner.

Toward the end of the 2000s and early 2010s, the video communications race changed once again. This time the race was focused on who could provide the best video communication experience. The idea was to use the best audio and video products on the market to create an environment that was indistinguishable from sitting in the same room with the people on the other end of the call. In late 2006, Cisco launched the CTS3000 Telepresence room system. *Telepresence* was the term first used here to describe the superior quality of the video call. This system supported three screens, three cameras, and three endpoints that all worked together to provide a life-sized video presence that almost gave the sensation of being in the same room as the participants connected in the call. Unfortunately, the CTS3000 system had one major drawback. For it to work the way Cisco intended, the company created a proprietary multiplexing protocol that compressed the RTP and RTCP streams. This meant that the only systems the CTS3000 could call were other CTS3000s. In 2008, Tandberg launched a telepresence room system called the T3, which stood for Tandberg 3-screen. This room system looked like something out of a science fiction movie. In fact, many people referred to it as the bridge on the Starship Enterprise from *Star Trek*. The T3 was far more advanced than the Cisco CTS3000 system, and it had the capability to call any other video communication device, including Cisco's flagship product, the CTS3000. Less than six months later, Tandberg launched another product called the Telepresence Server. This server could host multipoint calls for telepresence endpoints, and it could act as a gateway between the Cisco CTS3000 and other video communications devices. It was obvious to Cisco that the company was outmatched, so by October 2009 Cisco made an offer to acquire Tandberg. The vote was approved by Tandberg shareholders in December 2009 to sell to Cisco for \$3.4 billion.



Things got really interesting after that. Several other video communication companies protested the sale of Tandberg to Cisco. So, the first ever video telecommunications summit was held. Several changes were decided upon during that meeting. First, some terminology was officially canonized.

Devices capable of high-quality video communication would now be referred to as *telepresence endpoints*. The enhanced room systems, such as the T3 and CTS3000, would be referred to as *immersive telepresence endpoints*. The most significant change, however, was that the proprietary multiplexing protocol that Cisco created was offered as an open-source protocol that any vendor could use, and it was named the Telepresence Interoperability Protocol (TIP). By May 2010, Cisco closed on the acquisition of Tandberg and got busy to make use of the solutions the company now had at its disposal. In the years that followed, Cisco has provided telepresence innovations that have driven the growth of the industry. Some of the innovations include One Button to Push (OBTP), which makes connecting to a meeting as simple as pressing one button; Smart Scheduling; rendezvous conferencing; and now the next big change in the video telepresence race, cloud and hybrid collaboration.

Many companies began transitioning to the cloud several years ago. Microsoft announced the Office 365 platform, which is cloud-based. Life-Size, another video telepresence company that historically focused on endpoints exclusively, began offering infrastructure services in the cloud to which its endpoint product line could register. Newer companies were established that were centered around a cloud-based telepresence. Even data centers are now reaching out to the cloud, such as Amazon's AWS, Microsoft's Azure, and Google Cloud. It was not hard for Cisco to see the writing on the wall as to where the industry was heading. Cisco already had the Webex conferencing solutions, which are cloud-based. In an attempt to go beyond what Webex could offer, in 2013 Cisco came out with a product called Cisco Project Squared. After about a year of development, Cisco introduced some changes to this new cloud offering and changed the name to Cisco Spark. This platform ran for four years and continuously evolved into a very powerful cloud-based tool that companies could use for VoIP phone and telepresence endpoint registration and calling, instant messaging, document sharing, and multipoint conference meetings. Then in May 2018, Cisco changed the name once again to unify Spark with the Webex platform and called the new application Webex Teams. In 2019, Cisco combined a cloud calling platform it had acquired called BroadSoft to the Webex platform, creating the new Webex Calling solution. This new calling solution was much more robust, offered many more features, and could scale to any size company.

This actually created a problem. There were now three services called Webex that each performed a different function. There was the original *Webex Meetings*, which had its own app and web interface. There was *Webex Teams*, formerly Cisco Spark, which had its own app for messaging and used the Webex Control Hub as the web-based interface. Then there was *Webex Calling*, formerly BroadSoft, which used its own calling app and web interface. Cisco engineers worked tirelessly to unify these three services under a single pane of glass, as they call it. In 2021, the answer came in the release of a new application simply called the Webex app, which can be used for calling, meetings, and messaging. The term *Webex Teams* was retired, and all administration of these three services can be leveraged through the Webex Control Hub web interface.



At the same time that Webex Calling was being introduced to the Webex solution, Cisco was also developing a new concept that has once again changed the face of telepresence. Cognitive collaboration incorporates artificial intelligence (AI) into the collaboration products to help users in various aspects of their job. Webex is now interactive, much like Amazon Alexa, where you can talk to your telepresence endpoint registered to Webex in the cloud, and it will provide information in response or call a participant at your vocal command. If employees of a company use Webex Meetings to communicate with people outside their organization, a new tool called People Insight will provide background information on the other participants connected so that employees can know more about the people they're connecting with. Cisco is continually researching other cognitive tools to incorporate into the user experience. Webex was the communications solution on board the Artemis space shuttle launched by NASA in 2022 to return to the moon. Cisco is also working with some automobile companies to utilize Webex with the cars' internal systems. Clearly, communication has gone beyond simple voice and video. There are so many ways to communicate and share information. Collaboration involves using all the tools at your disposal to work together with other people. This is why Cisco calls its products *collaboration*.

What new innovations will the future hold? Will 3D projection become a reality, such as with Princess Leia in the Star Wars movie, *Episode IV—A*

New Hope? Perhaps 3D printing will be incorporated into these solutions in some capacity. Interactive automobiles and video walls in homes could become commonplace in the future. Android assistants could be used in corporations for various tasks. If history has proven anything, it is that people are limited only by imagination. If you can dream it, it can become reality.

Exam Preparation Tasks

As mentioned in the section “How to Use This Book” in the Introduction, you have a couple of choices for exam preparation: the exercises here, [Chapter 35](#), “[Final Preparation](#),” and the exam simulation questions in the Pearson Test Prep Software Online.

Review All Key Topics

Review the most important topics in this chapter, noted with the Key Topics icon in the outer margin of the page. [Table 1-2](#) lists a reference of these key topics and the page numbers on which each is found.



Table 1-2 Key Topics for [Chapter 1](#)

Key Topic Element	Description	Page Number
Paragraph	ISDN and POTS Explained	7
Paragraph	VoIP Explained	7
Paragraph	Video Telepresence Applications	9
Paragraph	IM and Presence Explained	10
Paragraph	UC Explained	10
Paragraph	Telepresence, Immersive Telepresence, and TIP Defined	12

Key Topic Element	Description	Page Number
Paragraph	Cognitive Collaboration Explained	13

Define Key Terms

Define the following key terms from this chapter and check your answers in the glossary:

B2B
 B2C
 H.320
 H.323
 IETF
 IM
 ISDN
 ITU
 OBTP
 PBX
 POTS
 Presence
 SIP
 TCP/IP
 Telepresence
 TIP
 UC
 VoIP

Q&A

The answers to these questions appear in [Appendix A](#). For more practice with exam format questions, use the Pearson Test Prep Software Online.

1. What product was invented in the late 1990s that is used to control VoIP phone?
2. What are the three communications protocols used for calls with video endpoints?
3. What were the three non-real-time communication services discussed in this chapter?
4. What are the two cognitive collaboration integrations with Webex discussed in this chapter?

Chapter 2

Audio Basics

This chapter covers the following topics:

Basic Understanding of Sound: This topic will introduce a basic understanding of how sound behaves, including wave propagation, technical properties of sound waves, and noise.

Analog vs. Digital Signals: This topic will compare noise on an analog signal compared to noise on a digital signal, observe the principle of the Nyquist-Shannon theorem, and identify how bandwidth conservation can be achieved through data compression.

ITU Audio Encoding Formats: This topic will identify the most commonly used audio codecs in a Cisco solution and analyze the various aspects of each codec.

The preceding chapter provided a backward look into the evolution of communications and a high-level overview of the technologies that exist today that allow us to collaborate at many levels. This chapter will begin to dive into the intricate details that encompass the vast world of audio communication. As you read through this chapter, you will find that the chasm that embodies audio communication is only a foot wide but a mile deep. Although this chapter will not cover every aspect of audio communication, it will provide a solid foundation to build your knowledge on. Topics discussed in this chapter include the following:

- **Basic Understanding of Sound**
- **Wave Propagation**
 - **Technical Properties of a Sound Wave**
 - **Understanding Noise**